Mixed-prime polyhedra from periods and dihedral extensions

human_compression project — private author draft

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1. The complete principal contract

Hou, Feng and Leemans [HFL] study regular polyhedra whose automorphism groups have order $2^n p$, where p is an odd prime. This application recovers their principal type restriction, both families of cubic constructions, their general odd-factor construction, and the exact existence range for type $\{6, 2^s\}$. We use their arXiv:2001.02945v2 edition, dated 14 July 2024, comprising seventeen pages. The correction in Section 6 concerns an auxiliary presentation in that edition; no claim about the later journal display is made.

We work with marked rank-three string C-groups. The distinguished generators a, b, c are involutions, $ac = ca$, and $\langle a, b\rangle \cap \langle b, c\rangle = \langle b\rangle$. This last equality is the complete rank-three intersection test. We retain the standard correspondence with regular abstract polyhedra, under which the group acts regularly on flags and its order is the number of flags. Nondegeneracy means both adjacent product orders are at least three. Conjugation is $x^g = g^{-1}xg$.

Theorem 1 (type restriction; [HFL, Theorems 1.2 and 3.1]). If a group of order $2^n p$ is generated by three elements of order at most two with commuting endpoints, then p divides at least one adjacent product order. Consequently, up to duality, a nondegenerate regular polyhedron of this order has type $\{2^s p, 2^t\}$ or $\{2^s p, 2^t p\}$.

Theorem 2 (odd-factor extension; [HFL, Theorem 4.1]). For positive odd integers ℓ_1, ℓ_2, integers $s, t \geq 2$ and $n \geq s + t + 1$, the explicit presentation in Section 3 is a solvable string C-group of type $\{2^s\ell_1, 2^t\ell_2\}$ and order $2^n\ell_1\ell_2$. In particular it gives the source's Corollary 4.2 by taking $\ell_1 = p$, $\ell_2 = 1$.

Theorem 3 (the exact six-sided range; [HFL, Theorem 1.3]). For $n \geq 7$, a nondegenerate regular polyhedron of order $3 \cdot 2^n$ and type $\{6, 2^s\}$ exists if and only if

$$2 \leq s \leq n - 2, \qquad s \neq n - 3. \tag{1}$$

The constructions identify the source's actual presentations, with the single sign correction specified in Section 6.

Theorem 4 (the cubic families; [HFL, Theorems 4.3 and 5.1]). The three literal presentations of type $\{6, 4\}$ give solvable groups of orders $96m^3, 192m^3, 384m^3$, for every integer $m \geq 1$. The source's three type-$\{6, 6\}$ presentations give solvable groups of orders $192m^3, 384m^3, 768m^3$, for every $m \geq 1$. In particular, type $\{6, 4\}$ and type $\{6, 6\}$ both occur at order $3 \cdot 2^n$ for every $n \geq 5$.

The shared ingredients are the supported-kernel and square-group results of [UA], and the complete affine-lattice criterion of [AL]. The existing exact $\{6,6\}$ calculation [C66] is reused. Burnside's p^aq^b solvability theorem and Lucchini's cyclic-core bound are retained classical inputs, as in the source. One bounded exclusion retains the complete GAP SmallGroups catalogue of order 384. These inputs, rather than experimental extrapolation, are the precise boundaries of the recovery.

2. A dihedral quotient gives the type restriction

Lemma 5. If a finite group G of order 2^np is generated by involutions, then $G/O_2(G) \cong D_{2p}$, where $O_2(G)$ is its largest normal two-subgroup.

Proof. Put $\overline{G} = G/O_2(G)$. It still has an odd part of order p, is generated by involutions or identities, and has no nontrivial normal two-subgroup. Burnside's theorem makes it solvable. A minimal normal subgroup is therefore elementary abelian, and must have order p. Write it as P.

Its centralizer $C = C_{\overline{G}}(P)$ is normal. If S is a Sylow two-subgroup of C, then $C = P \times S$: P is central in C and the orders multiply to $|C|$. Thus S is the unique Sylow two-subgroup of C, hence characteristic there and normal in $\overline{G}$. It is trivial. Therefore $C = P$ and $\overline{G}/P \hookrightarrow \mathrm{Aut}(P)$ is cyclic. Because it is generated by elements of order at most two, this quotient has order at most two. It cannot be trivial: the nontrivial group of odd order P cannot be generated by involutions. Its order is two and its action on P is the unique nontrivial involution of $\mathrm{Aut}(C_p)$, namely inversion. A generating involution outside P supplies a complement. Hence $\overline{G} = D_{2p}$. □

Suppose neither adjacent product has order divisible by p. Their images in D_{2p} have order at most two, and the endpoint product also has order at most two. Three involutions or identities with all pair products of order at most two commute pairwise. They would generate an abelian two-group, not D_{2p}. This contradiction proves Theorem 1. The allowed type forms follow from Lagrange's theorem. The exponents s, t in this type restriction are not silently restricted to those of the narrower construction in Theorem 2.

3. Two independent edge periods

Write $h = 2^s\ell_1$, $k = 2^t\ell_2$ and $d = n - s - t \geq 1$. Put

$$R = \{a^2, b^2, c^2, (ab)^h, (bc)^k, (ac)^2, [(ab)^4, c], [a, (bc)^4]\}.$$

The exact source presentation is

$$G = \begin{cases} \langle a, b, c \mid R, [(ab)^2, c]^{2^{(d-1)/2}} \rangle, & d \text{ odd}, \\ \langle a, b, c \mid R, [(ab)^2, (bc)^2]^{2^{(d-2)/2}} \rangle, & d \text{ even}. \end{cases} \tag{2}$$

The square-group results UA.Affine and UA.Square identify the quotient obtained by adding $(ab)^4 = (bc)^4 = 1$. It is the marked affine square group modulo the lattice

$$\Lambda = \begin{cases} 2^{(d+1)/2}\mathbb{Z}^2, & d \text{ odd}, \\ 2^{d/2}\langle (1,1), (1,-1) \rangle, & d \text{ even}. \end{cases} \tag{3}$$

Call it B. It has order 2^{d+4}, type $\{4,4\}$, IP, and every Boolean deletion. The least scale in (3) is two, including $d = 1, 2$; there is no unverified small seed. These precise presentations and their normal forms are part of the retained umbrella, not inferred from their orders.

Apply UA.SK to the adjacent pairs, using the dihedral quotients $D_{2h} \to D_8$ and $D_{2k} \to D_8$. The resulting compatible product H has type $\{h, k\}$, IP and order

$$|H| = 2^{d+4}(h/4)(k/4) = 2^n \ell_1 \ell_2. \tag{4}$$

The relators in (2) hold in it. For the final relator, the base coordinate is killed by (3), and every attached coordinate is trivial: in either dihedral factor, one of the commutator entries is trivial or the relevant square is the identity. The fourth-power commutators also vanish in every coordinate. Thus the presented group maps onto H.

For the reverse bound, $A = \langle (ab)^4 \rangle$ is normal in G and has order at most $h/4$; its normality follows from inversion by a, b and the displayed commutation with c. In G/A, the cyclic subgroup on $(bc)^4$ is normal, of order at most $k/4$. Killing it gives exactly B by the presentation. These two finite cyclic layers bound $|G|$ by the right side of (4). The surjection attaining the bound is therefore an isomorphism of marked groups. This proves Theorem 2, including its literal presentation. Solvability follows from the abelian-by-dihedral square group and the two cyclic layers.

4. One affine group for the three {6,4} families

Define integral affine transformations on $\mathbb{R}^3$ by

$$a(x, y, z) = (y, x, z), \quad b(x, y, z) = (1 - x, z, y), \quad c(x, y, z) = (x, y, -z). \tag{5}$$

They satisfy the complete source relators

$$U = \langle a, b, c \mid a^2, b^2, c^2, (ab)^6, (bc)^4, (ac)^2, [a, (bc)^2]^2 \rangle. \tag{6}$$

Their linear parts are signed permutation matrices. On the signed basis $(e_1, e_2, e_3, -e_1, -e_2, -e_3)$ they act as

$$(1\ 2)(4\ 5), \quad (1\ 4)(2\ 3)(5\ 6), \quad (3\ 6). \tag{7}$$

They generate the full signed permutation group Q, of order 48: the coordinate permutations generate S_3, and conjugates of the last sign change generate all coordinate sign changes. The exact Schreier transition calculation of AL.Lemma 1 gives translation kernel

$$L = 2\mathbb{Z}^3 + \mathbb{Z}(1, 1, 1), \qquad [\mathbb{Z}^3 : L] = 4. \tag{8}$$

For example the transitions include vectors generating $2\mathbb{Z}^3$ and an all-odd vector; every transition has three coordinates of the same parity. These two containments prove (8). A separate exact Reidemeister–Schreier/Tietze computation for the kernel of the marked map (6) to (7) returns three generators with the three pairwise commutator relations. Its actual presentation is free

abelian of rank three. AL.Lemma 2 therefore identifies the full presentation (6) with the affine group (5). The computation proves a statement about the abstract kernel, not just about its abelianization.

Here are the source's three triples of translation words. We keep them because matching their actual values is necessary to recover the source presentations.

$$\begin{array}{lll} x_1 = (bc^{ba})^2, & y_1 = (c^b b^a)^2, & z_1 = (cb^{ab})^2, \\ x_2 = (a^b b^{ac})^2, & y_2 = (b^{acb} a)^2, & z_2 = (a^{bcb} b^{ab})^2, \\ x_3 = (bac)^6, & y_3 = (cba)^6, & z_3 = ((cba)^b)^6. \end{array} \tag{9}$$

Their translation vectors are, in the order x_i, y_i, z_i,

$$\begin{aligned} i = 1: &\quad (2,0,0),\ (0,-2,0),\ (0,0,-2), \\ i = 2: &\quad (2,0,2),\ (2,-2,0),\ (-2,-2,0), \\ i = 3: &\quad (2,-2,2),\ (2,-2,-2),\ (-2,-2,-2). \end{aligned}$$

The invariant period lattices and their indices are

Triple	Lattice Λ_i	$[L : \Lambda_i]$
1	$2\mathbb{Z}^3$	2
2	$2D_3$	4
3	$4\mathbb{Z}^3 + 2\mathbb{Z}(1,1,1)$	8

Here $D_3 = \{(u,v,w) \in \mathbb{Z}^3 : u+v+w \text{ is even}\}$. Their basis determinants have absolute values $8, 16, 32$, respectively, and all three lattices are invariant under signed permutations. Imposing $x_i^m = y_i^m = z_i^m = 1$ gives exactly $U/(m\Lambda_i)$, by AL.Theorem 3. Its order is $96m^3, 192m^3$ or $384m^3$.

The adjacent affine subgroups have orders twelve and eight, and inject into their linear images. The latter meet only in 1 and the linear part of b. This is checked on the finite point group and implies IP in every period quotient. The adjacent orders remain six and four. The point group $C_2^3 \rtimes S_3$ is solvable, so every quotient is solvable. This proves the full source Theorem 4.3. Taking $m = 2^j$ gives exponents $5+3j, 6+3j, 7+3j$, proving its entire Corollary 4.4 without three separate atlas identifications.

The matrix convention is explicit. The affine products in (5) act on the rightmost factor first. GAP uses right point actions; a point permutation corresponds to the inverse column-action matrix. Since the three generating matrices are involutions, their displayed permutations agree with (7), and the kernel calculation is unchanged.

5. All extensions except one seed

Set $x = bc$. The abstract group (6) has a retraction onto D_8 obtained by killing a. The three triples (9) then become

$$(x^{-2}, x^2, x^{-2}), \qquad (1,1,1), \qquad (x^6, x^{-6}, x^6). \tag{10}$$

Consequently the retraction descends to $U/(m\Lambda_1)$ and $U/(m\Lambda_3)$ when m is even, and to $U/(m\Lambda_2)$ for every m. This is the precise reason the first and third minimal seeds cannot be used in this extension. The other pair deletions also hold, but only the displayed retraction is needed.

For $s \geq 2$, replace the second dihedral group by $D_{2^{s+1}}$ in the compatible product of AL.Theorem 4. It multiplies the seed order by 2^{s-2}, preserves the first adjacent group, and gives type $\{6, 2^s\}$. Write $d = n - s$. With $m = 2^j$, the three families cover

$$d = 3 + 3j \ (j \geq 1), \qquad d = 4 + 3j \ (j \geq 0), \qquad d = 5 + 3j \ (j \geq 1). \tag{11}$$

These are precisely all $d \geq 4$ except $d = 5$.

The source's Lemma 4.6 supplies the latter case. Let

$$w = bacba(cb)^2a, \qquad V = \langle a, b, c \mid a^2, b^2, c^2, (ab)^6, (bc)^4, (ac)^2, w^2 \rangle. \tag{12}$$

An exact finite presentation computation gives $|V| = 384$, the adjacent orders six and four, and all sixty-four subset intersections. This is an explicit checked seed, not an imported atlas entry. Killing a defines the required retraction: w becomes $(bc)^3b$, a reflection in every dihedral group. Therefore

$$V_s = \langle a, b, c \mid a^2, b^2, c^2, (ab)^6, (bc)^{2^s}, (ac)^2, [a, (bc)^4], w^2 \rangle \tag{13}$$

is exactly its compatible product with $D_{2^{s+1}}$, by the exact presentation variant of AL.Theorem 4. It has order $384 \cdot 2^{s-2} = 3 \cdot 2^{s+5}$ and type $\{6, 2^s\}$ for every $s \geq 2$. This is the entire source Lemma 4.6, with $n = s + 5$. The new seed and the third family of (9) are not interchangeable merely because both have order 384 at their first parameter: only the former has the required deletion at that parameter.

6. Exact period presentations and a corrected sign

For completeness we identify the source's Lemma 4.5 presentations, rather than merely produce the same types and orders. Put

$$R_s = \{a^2, b^2, c^2, (ab)^6, x^{2^s}, (ac)^2, [a, x^2]^2, [a, x^4]\}.$$

For $d = n - s \geq 4$, $d \neq 5$, use the following extra relations:

d mod 3	M	period relations
0	$2^{(d-3)/3}$	$x_1^M = x^{-2M}, \quad y_1^M = x^{2M}, \quad z_1^M = x^{-2M},$
1	$2^{(d-4)/3}$	$x_2^M = y_2^M = z_2^M = 1,$
2	$2^{(d-5)/3}$	$x_3^M = x^{6M}, \quad y_3^M = x^{-6M}, \quad z_3^M = x^{6M}.$

(14)

The first exponent is negative in (14). It is positive in the literal v2 Lemma 4.5, physical page 12. All other entries agree with that source. In the first and third branches M is even; in the middle branch $M \geq 1$.

Here is a presentation proof valid for all parameters. Start with the presentation R_s alone. The cyclic subgroup $C = \langle x^4 \rangle$ is normal, has order at most 2^{s-2}, and its quotient is (6). The affine group (6) retracts to D_8, so the presentation maps onto its compatible product with $D_{2^{s+1}}$. The kernel bound and the exact infinite-base version of AL.Theorem 4 identify the two groups. In this product, each extra relation (14), using (10), is precisely $(mv, 1) = 1$ for one of the three source translation vectors v. Normal closure kills exactly $m\Lambda_i \times 1$. The lattice lies in the retraction kernel by the parity condition already checked. Hence the quotient is exactly the finite compatible product of Section 5. This proves its order, marking and IP with no unproved extension consistency.

The sign correction is forced by an elementary identity, independent of the affine calculation. For any involutions a, b, c, the words (9) satisfy

$$x_1^a = y_1^{-1}. \tag{15}$$

Indeed $x_1 = (babcba)^2$, $y_1 = (bcbaba)^2$, and conjugation by a reverses the latter word. In the first branch the literal source imposes both $x_1^M = x^{2M}$ and $y_1^M = x^{2M}$. Since M is even and $[a, x^4] = 1$, conjugating the first equation by a and comparing with the inverse of the second yields

$$x^{4M} = 1. \tag{16}$$

For $n = 10, s = 4$, one has $M = 2$. Thus the literal presentation cannot have the advertised rotation order sixteen. GAP computes order 1536 and rotation order eight for that presentation. With the corrected first sign, its order is 3072, its rotation order is sixteen, and all intersections hold. Three Lean declarations prove (15), the general period-sign implication, and (16) directly from the literal conjugation words. They do not rely on this numerical example.

This is a correction to an auxiliary construction in the bound edition, not a failure of Theorem 3. The corrected family proves the required existence statement for every parameter. It is not reported as a solved open conjecture.

7. The excluded excess and the full existence theorem

The rank-three product bound gives $2 \cdot 6 \cdot 2^s \leq 3 \cdot 2^n$, so $s \leq n - 2$; nondegeneracy gives $s \geq 2$. If $s = n - 2$, the standard tight construction for two even adjacent orders follows from UA.SK applied to the rank-three Boolean group. It has exactly $3 \cdot 2^n$ elements. Sections 5 and 6 supply every $s \leq n - 4$. It remains to exclude $s = n - 3$.

We retain Lucchini's cyclic-core theorem [L98, Proposition, p.241]: for a proper cyclic subgroup A of index t in a finite group G, $[G : \mathrm{core}_G(A)] \leq t^2 - t$. Dividing by t gives $[A : \mathrm{core}_G(A)] < t = [G : A]$. The complete original paper was read and is bound as a retained primary; its proof is not rederived here. In a hypothetical group of order $3 \cdot 2^n$ and type $\{6, 2^{n-3}\}$, the second rotation subgroup $A = \langle bc \rangle$ has index twenty-four. For $n \geq 8$, its order is at least thirty-two, so its core is nontrivial. Its unique subgroup N of order two is characteristic in the cyclic core and therefore normal in G.

The quotient G/N preserves the first dihedral subgroup injectively: that subgroup meets $\langle b, c \rangle$ in $\langle b \rangle$, whereas N lies in the rotation subgroup. Because $N \leq \langle b, c \rangle$, the adjacent intersection

in the quotient is still exactly the image of $\langle b\rangle$. To see this directly, if an element of the first subgroup maps into the second one, it already lies in that second subgroup, since its full preimage is unchanged by adjoining N. Thus the quotient is a string C-group of order $3 \cdot 2^{n-1}$ and type $\{6, 2^{n-4}\}$. Descent reaches $n = 7$.

For that base, an independent exhaustive calculation uses the retained complete SmallGroups catalogue of groups of order 384. For every group G it first tests whether its exponent is divisible by sixteen. For each remaining group, it takes one representative c from every conjugacy class of involutions. It enumerates every involution b with $|bc| = 16$, modulo the conjugation action of $C_G(c)$. For each such pair it takes every involution $a \in C_G(c)$ with $|ab| = 6$, and tests whether $\langle a, b, c\rangle = G$.

Every possible marking is included: simultaneous conjugation can first place c at its representative and then move b within its centralizer orbit; the exhaustive centralizer search still contains the corresponding a. The complete computation visits 20,169 catalogue groups, including 1,178 whose exponent passes the first test, and 4,436 reduced adjacent pairs. There is no generating triple, even before imposing IP. Thus no required C-group exists at order 384. This is a catalogue-based computer-assisted proof, not an unexplained statement that a software package returned no example. The algorithm and counts are replayable. The induction proves the exclusion for all $n \geq 7$, completing Theorem 3 without a residual list of unspecified small cases.

8. The reused {6,6} calculation

The already proved component [C66] uses

$$a(x, y, z) = (y, x, z), \quad b(x, y, z) = (1 - x, z, y), \quad c(x, y, z) = (-y, -x, z). \tag{17}$$

Its point quotient is the same signed permutation group of order 48; its translation kernel is $2D_3$, of index sixteen in $\mathbb{Z}^3$. The retained arithmetic checker records raw affine differences. Its translation-lattice conclusion uses the invariant-lattice version of AL.Lemma 1: $2D_3$ is signed-permutation invariant, contains every raw difference, and is generated by exhibited raw differences. No general claim that arbitrary raw differences alone generate a kernel is used. An exact index48 kernel presentation identifies the complete universal group

$$\langle a, b, c \mid a^2, b^2, c^2, (ab)^6, (bc)^6, (ac)^2, (cbab)^3\rangle \tag{18}$$

with this affine group. The actual source period words give the lattices $4\mathbb{Z}^3$, $4D_3$ and $8\mathbb{Z}^3 + 4\mathbb{Z}(1, 1, 1)$, of indices four, eight and sixteen in $2D_3$. The cited component contains their word-by-word evaluations; those are part of this application's dependency, not omitted obligations. AL.Theorem 3 gives the orders $192m^3, 384m^3, 768m^3$.

Both adjacent parabolics have order twelve and faithful linear image. Their linear intersection has four elements. Two lift to the common identity and middle involution; the other two differ by translation $(0, 2, 2)$. None of the three period lattices, or their positive multiples, contains that vector, since they lie in $4\mathbb{Z}^3$. This is the complete all-parameter IP check. The quotients are solvable by the same point-group argument as before. It proves the exact source Theorem 5.1, with its presented groups, rather than another family sharing their numerical parameters.

Taking $m = 2^j$ covers all exponents $n \geq 6$ in order $3 \cdot 2^n$. For $n = 5$, take the tetrahedral marking $(12), (23), (34)$ in S_4, and attach independent central involutions u, v to the first and third markers. The resulting marking in $S_4 \times \langle u, v \rangle$ has adjacent products of order six. Their cubes recover u, v, so it generates all 96 elements. Its two parabolic kernels are respectively $\langle u \rangle$ and $\langle v \rangle$; the kernel criterion preserves the tetrahedral intersection. This proves the full source Corollary 5.2. The proof and its attribution are reused from [C66].

9. Scope, attribution and verification

All four principal results remain attributed to Hou–Feng–Leemans. The new contribution here is the explicit second affine model, the shared period-lattice proof, the exact presentation identification of the extensions, and the independently reproduced nonexistence base. The source's cyclic-core descent is retained with its hypotheses and quotient step made explicit. The mixed-prime type argument is a short reworking of its solvable-group argument. Neither reuse nor a shorter presentation is a claim to the discovery of these established results.

The new common paper [AL] and the retained umbrella [UA] carry genuine proof obligations. The cubic point-kernel computations and the order384 catalogue search are exact finite computations with declared external algorithms. The three new Lean declarations prove the literal sign obstruction; they do not establish the entire source contract in Lean. The previously checked kernel and supported-extension interfaces remain available, with their own stated scope. All programs, retained input statements and umbrella costs must be included in later matched measurements. No numerical rate is asserted by counting paper titles or pages alone.

The source's broader Problem 1.1, the characterization of all groups of order 2^n or $2^n p$ arising from regular or chiral polytopes, is not solved here. Its final question about type (2) for primes at least five has already been answered in the literature, as recorded in the project's question audit and the recovered equal-label construction [EL]. That existing answer earns no new question credit here. Introductory survey statements, the small-order table as an independently complete census, and ancillary uniqueness remarks about the two order384 seeds are outside this principal contract.

References

[HFL] D.-D. Hou, Y.-Q. Feng and D. Leemans, *Regular 3-polytopes of order* $2^n p$, arXiv:2001.02945v2, 14 July 2024, 17 pp. The later journal publication is not the edition collated in this application.

[AL] *Period lattices and marked affine quotients*, `phase250_affine_lattice_umbrella.md`.

[UA] The frozen Topic A umbrella, `phase207_topic_A_umbrella.md`, UA.K, UA.SK, UA.Affine, UA.Square, UA.Edge. Reuse does not make an entire umbrella free in a topic's comparison.

[C66] *One affine model for all three {6,6} lattice families*, `phase2_cubic_lattices.md`, with `phase2_cubic_lattice_check.py` and `phase2_cubic_kernel_check.g`. This retained proof includes all source word evaluations and its original attribution.

[EL] *Even subsets of a polygon and an equal-label presentation*, `phase229_equal_label_application.md`, recovering the polygon construction of Montero–Ivić Weiss and the exact later presentation of Li–Zhang.

[L98] A. Lucchini, *On the order of transitive permutation groups with cyclic point-stabilizer*,

Rendiconti Lincei, Matematica e Applicazioni, series9, volume9 (1998), 241–243, Proposition on p.241. The original primary is `sources/phase254_Lucchini1998.pdf`; its two archive cover pages precede the three article pages. The theorem is also restated in [HFL, Proposition2.4], whose secondary reference is not needed here.

The retained solvability theorem is the classical Burnside $p^a q^b$ theorem. The finite classification input is the complete GAP SmallGroups catalogue at order 384, not a classification proved in this application.